

Three choices, three different questions

1

Fracture formulation

What represents separation?

A phase-field energy or a cohesive traction-separation law.



discretise the chosen equations

2

Spatial discretisation

How are the fields approximated?

Standard finite elements, XFEM enrichment, mesh resolution.



solve the discrete equations

3

Solution algorithm

How is the discrete problem solved?

Staggered or monolithic updates; Newton or quasi-Newton.

Choose a fracture model, spatial approximation and solution algorithm.